

Set up with uv

For students without Python or Anaconda. Windows, macOS and Linux.

Before you start - pilot edition

Use the instructor-issued course folder containing **pyproject.toml**, **.python-version** and **uv.lock**. If any are missing, obtain the complete folder before continuing. The uv files and lockfile are not yet included in the current repository.

You need an internet connection for the first setup. uv downloads Python and the required packages; your existing R installation can stay as it is. Use this route or the Anaconda route for the course.

1 Install uv once

Windows: open PowerShell and run:

```
winget install --id=astral-sh.uv -e
```

macOS / Linux: open Terminal and run:

```
curl -LsSf https://astral.sh/uv/install.sh | sh
```

Close and reopen the terminal, then run **uv --version**. If WinGet is unavailable, use the Windows installer at docs.astral.sh/uv/getting-started/installation/.

2 Open the course folder

Extract the downloaded course archive first. In your terminal, enter the folder containing pyproject.toml. Replace the example path with your own:

```
cd "path/to/ESBMTK-practicals"
```

Type commands into the terminal, not into RStudio or a notebook cell. Run one line at a time; wait until the prompt returns.

3 Create the course environment

```
uv sync --locked
```

Wait for the installation to finish. uv creates a **.venv** folder and uses the package versions selected by your instructor. Keep this folder inside the course folder.

Open your notebooks

uv route / finish setup, check it, and return next time

4 Register the kernel and launch JupyterLab

From the same course folder, run these two commands. Kernel registration is a one-time step:

```
uv run --locked python -m ipykernel install --sys-prefix --name esbmtk-practicals
uv run --locked jupyter lab
```

JupyterLab opens in your browser. Open **notebooks/student/**. Select **esbmtk-practicals** in the kernel selector, including if the notebook asks for the instructor's ESBMTK314 kernel. A kernel is the Python process running your cells.

Run a quick check

In JupyterLab, create a new Python notebook with the course kernel. Paste this into a cell and press **Shift+Enter**:

```
import sys
import numpy as np
import esbmtk
import PyCO2SYS

print(sys.executable)
np.testing.assert_allclose(np.linalg.solve(np.eye(2), np.ones(2)), [1, 1])
print("Setup check passed")
```

Expect **Setup check passed**. The printed Python path should contain **.venv**. This checks imports and a real numerical calculation; it does not complete any course exercise.

Restart JupyterLab after shutting it down

Only when JupyterLab is no longer running (for example, after restarting your laptop), open a terminal in the course folder and run:

```
uv run --locked jupyter lab
```

If JupyterLab is already running, open existing notebooks in its file browser or create one via **File > New > Notebook**, then select the course kernel. **No terminal command is needed for each notebook**. Closing a browser tab usually leaves JupyterLab running. Save your work before shutting it down.

If something does not work

uv not found: reopen the terminal after installation.

Missing or outdated lockfile: ask for the complete course folder; keep `--locked`.

Wrong Python path or missing module: choose esbmtk-practicals and restart that kernel.

Exercise / NotImplementedError: a marked student placeholder is expected; follow the exercise.

Help: docs.astral.sh/uv/guides/integration/jupyter/. For other errors, send your instructor the complete error message and your operating system.